

NHL Career Curve for Forwards

A study of point production over time

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Abstract

The focus of this study was on NHL forwards with a minimum ten years service through the end of the 2005-06 regular season. Percentage of career to date points was examined by the age at which a player started his NHL career. Several age brackets were tested and, due to similarities in trends (or lack of a meaningful number of players), were collapsed into three groups—eighteen to nineteen, twenty to twenty-one and twenty-two or older. Overall, those players beginning their NHL careers at an older age tended to peak in terms of point production sooner than those who started at an earlier age.

Non-linear models were fit for each group to estimate the actual cumulative percentage of career to date points scored in each season. The first derivatives of these models were also used to arrive at a predicted rate of change to cumulative career to date points. For the older age group, that rate of change peaked after the third season in the League, coinciding with the expiration of typical entry-level contracts. Interestingly, for those in the two younger age brackets, the rate of change peaked in the fourth year. As with the actual results, those beginning their career later in life were predicted to slow down quicker as time elapsed.

It is important to keep in mind that the results of this study were based on the average NHL forward; that is, on group statistics. When looking at an individual, the non-linear models may not hold true. For example, Vincent Damphousse scored in an approximate linear manner over his 18-year career, as opposed to the overall curved shape that was exhibited by the group of players. That points to his relative consistency over a long time in the League. Again, *on average*, cumulative point production does not follow this nice, linear trend. At the individual level, caution must be taken when applying these generalized shapes, which might turn this paper into a purely academic exercise. However, it was worth examination and provided general trends that can be expected over the career of a forward in the NHL.

Data

Career data through the 2005-06 regular season was extracted from the NHL web site and segregated by position. Since the purpose of this study was to examine point production, only forwards were examined. As the time factor was also considered, the study was further limited to those with at least ten years of service. The lockout season of 2004-05 was not counted as a year

of service, though it did give some young NHLers time to hone their skills in the minor leagues (e.g., Corvo, Spezza ...).

In total, 160 players were included in the analysis, ranging from the minimum ten to 25 years of service. The starting age was computed, in months, as of October 1 of a player's first NHL season and was distributed as shown at right. The first season that a qualifying player started was 1979-80 and the latest one started was 1995-96. Those starting their NHL careers in 1995-96 contributed ten years of data; players starting in 1994-95 had eleven years and so forth.

Starting Age	#	%
[18, 20)	73	45.63
[20, 22)	57	35.63
[22, Older)	30	18.74
Total	160	100.00

We see that around half (48.75%) of the players began their NHL careers in 1992-93 or more recent, while only 19% began their career prior to 1989-90. The data spanned several seasons and was aligned according to season number for that individual player regardless of which year was his first. That is, if a player began his career in 1995-96 his point total for that year was considered his points for year number one. Similarly, if a player began his career in 1990-91, then that season was considered year one. The yearly points were then summed by age group and their cumulative percentages examined.

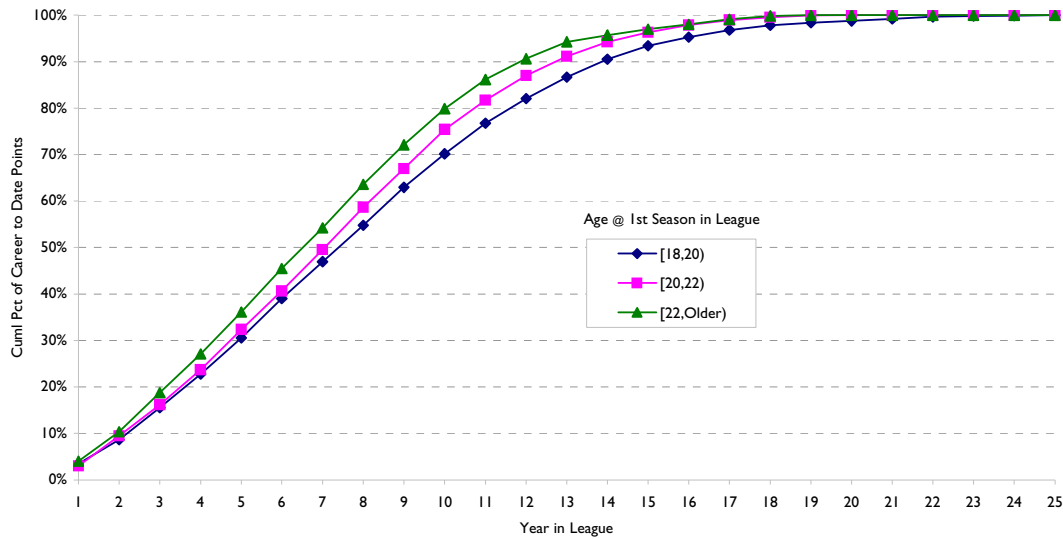
1st NHL Season	#	%	Cuml %
1995-96	19	11.88	11.88
1994-95	21	13.13	25.00
1993-94	22	13.75	38.75
1992-93	16	10.00	48.75
1991-92	24	15.00	63.75
1990-91	19	11.88	75.63
1989-90	9	5.63	81.25
1988-89	10	6.25	87.50
1987-88	4	2.50	90.00
1986-87	6	3.75	93.75
1985-86	4	2.50	96.25
1984-85	2	1.25	97.50
1983-84	1	0.63	98.13
1982-83	1	0.63	98.75
1981-82	1	0.63	99.38
1979-80	1	0.63	100.00
Total	160	100.00	

Below find the cumulative points distribution by age in the players' first NHL regular season; this formed the data for calculating the career curves by starting age.

Year	[18,20)		[20,22)		[22,Older)		Total	
	Points	Cuml %	Points	Cuml %	Points	Cuml %	Points	Cuml %
1	1,616	3.47	910	3.01	581	4.02	3,107	3.40
2	2,421	8.66	1,945	9.44	916	10.35	5,282	9.19
3	3,194	15.52	2,045	16.21	1,219	18.77	6,458	16.26
4	3,381	22.77	2,263	23.69	1,203	27.09	6,847	23.76
5	3,630	30.56	2,619	32.35	1,307	36.12	7,556	32.04
6	3,961	39.06	2,499	40.62	1,350	45.45	7,810	40.59
7	3,685	46.97	2,693	49.53	1,273	54.25	7,651	48.97
8	3,648	54.80	2,759	58.65	1,350	63.58	7,757	57.47
9	3,804	62.96	2,518	66.98	1,232	72.09	7,554	65.74
10	3,369	70.19	2,545	75.40	1,129	79.89	7,043	73.45
11	3,063	76.76	1,917	81.74	903	86.14	5,883	79.90
12	2,466	82.05	1,590	87.00	652	90.64	4,708	85.05
13	2,141	86.65	1,244	91.11	527	94.28	3,912	89.34
14	1,818	90.55	954	94.27	207	95.71	2,979	92.60
15	1,349	93.44	625	96.34	189	97.02	2,163	94.97
16	857	95.28	460	97.86	145	98.02	1,462	96.57
17	709	96.80	339	98.98	154	99.09	1,202	97.89
18	464	97.80	175	99.56	113	99.87	752	98.71
19	255	98.35	112	99.93	19	100.00	386	99.13
20	193	98.76	22	100.00			215	99.37
21	216	99.23					216	99.60
22	197	99.65					197	99.82
23	81	99.82					81	99.91
24	40	99.91					40	99.95
25	43	100.00					43	100.00
Total	46,601		30,234		14,469		91,304	

Methodology & Results

The first step was to graph the cumulative percentage of points scored by year and by age group to see what the shapes of the distributions looked like. It should be noted that several combinations of age groupings were examined before arriving at the age bands presented in this paper; the final bins followed a logical collapsing of groups exhibiting like values.



From the above illustration, it is apparent that point production varied with the starting age of a player. Those who started playing NHL hockey at 22 or older scored about 80% of their career points at ten years. At the same point in time (the 10-year mark), those who began their NHL careers at an age less than 20 were only at the 70% level; they hit the 80% level during their twelfth year. This means that, in terms of cumulative point production, those starting their NHL careers at an earlier age tended to have greater longevity. As well, those beginning later in life had a tendency to peak sooner.

Once the shapes of the curves above became visible, it was apparent that non-linear models would have to be fit. Several curve forms were attempted and the final models chosen were of the form:

$$f(x) = a + bx + c x^{1/2} + d \ln[x] \quad (1)$$

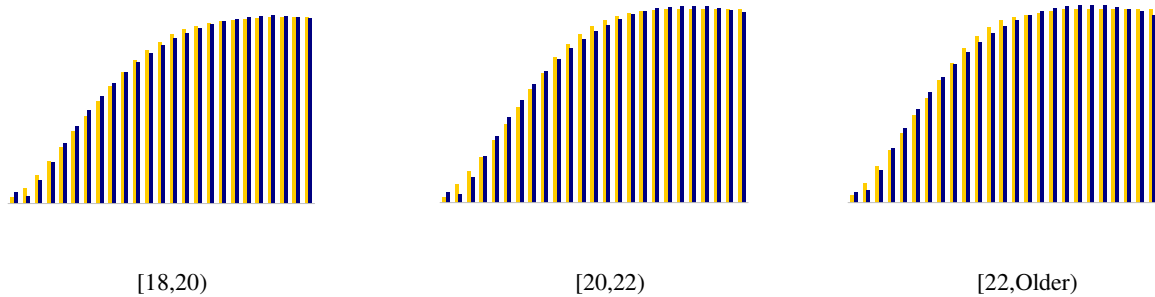
A separate model of the above form was fitted for each age group.

While the shape of each model was chosen to be consistent across all age groups, the parameter estimates of a , b , c and d were all different. Confidence intervals were computed for each estimate to see whether or not the value zero was contained in the interval; none contained zero, which means the parameter estimates were not likely to be zero.

The β estimates, approximate standard errors and 95% confidence intervals for each of the three models were as follows:

	[18,20)				[20,22)				[22,Older)			
	β est.	S.E.	95% CI		β est.	S.E.	95% CI		β est.	S.E.	95% CI	
a	-2.0258	0.1100	-2.2546	-1.7970	-2.2233	0.1298	-2.4932	-1.9534	-2.1562	0.1240	-2.4141	-1.8983
b	-0.1921	0.0117	-0.2164	-0.1678	-0.2166	0.0138	-0.2453	-0.1879	-0.2160	0.0132	-0.2434	-0.1886
c	2.2781	0.1308	2.0062	2.5501	2.4971	0.1543	2.1762	2.8179	2.4280	0.1474	2.1214	2.7345
d	-1.1086	0.0821	-1.2794	-0.9378	-1.2005	0.0969	-1.4020	-0.9989	-1.1221	0.0926	-1.3147	-0.9296

The **actual** versus **predicted** values for each year in the NHL (x -axis) were graphed for each model to give us a visual verification:

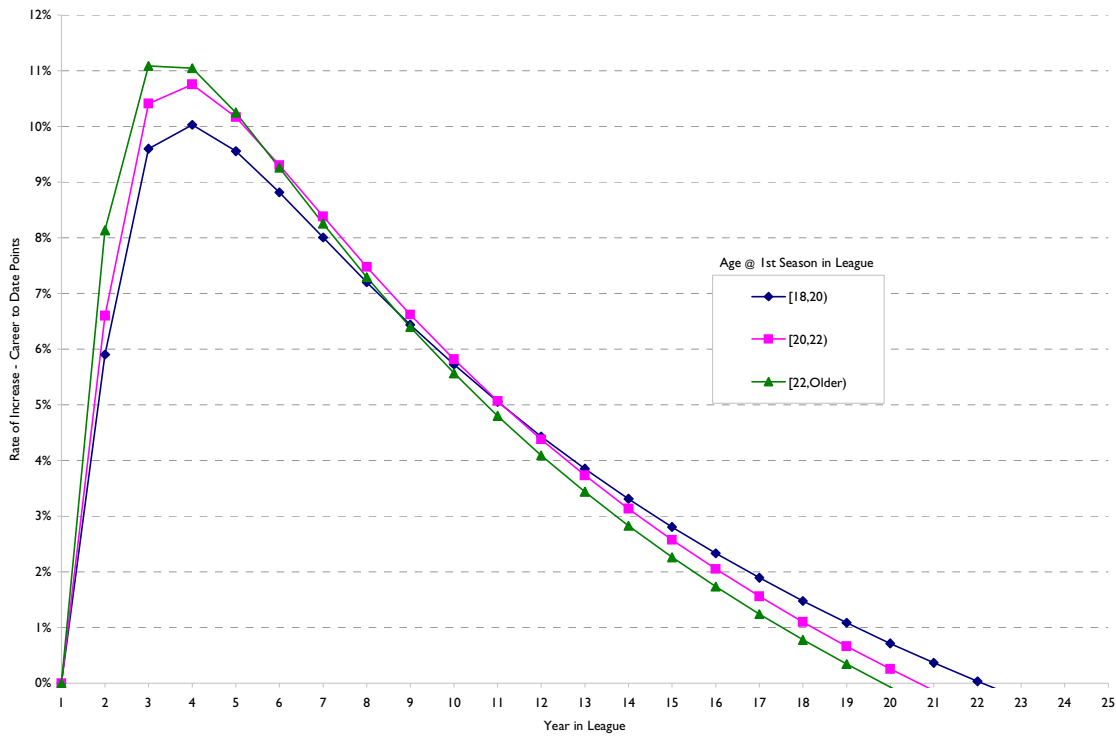


From the graphs above, we note that the models fit the data rather well. Also, the residuals for each model summed to zero, as was expected, another indication of appropriate models.

Having modeled shapes for each of the age groups enabled us to have a tool to apply to forwards entering the League at various stages of their lives. Taking the first derivative of Equation (1) allowed us to examine the rate of change in cumulative predicted points:

$$f'(x) = b + \frac{1}{2}c x^{-1/2} + \frac{d}{x} \quad (2)$$

It should be noted that the derivative value for each model at year one was negative, which is nonsensical. Therefore, the values for year one were set to zero.



As we examine the rate of increase in career to date percentage of points, we can see that the 22 and older age bracket peaked in terms of increasing predicted point production in their third NHL season, coincidentally the expiration time of many entry-level contracts, perhaps lending credence to the thought that certain players step in up in their contract years. Those in the other two age brackets added the most to their cumulative totals in their fourth year.

We also note that the younger age bracket was predicted to keep producing for a longer period of time than the older age groups. For example, in the fifteenth year, those who started at an age less than 20 added 2.8% of their predicted career points (this compares well to the actual value of 2.9%), while those who began at 22 or older scored 2.3% of their predicted career totals. As well, the younger-starting players lasted about three seasons longer.

Pitfalls & Caveats

The above approach to studying the career span of an NHL forward is not without its pitfalls. Some players played longer than others and they were all aggregated to give an average picture of the life span of an NHL career for a forward. A number of players are still active and continue to

accumulate points. Perhaps limiting the study to only retired players who played the same number of years would have made more sense, but that dataset would have been too small from which to draw any meaningful conclusions (recall that most of the players in the dataset began their careers in the 1990s). Era differences, while probably present, should have provided less a problem than the data quantity. Also of note is that few players made it to 20 years or more, thus rendering predictions at that late stage of a career uncertain.

These formulae should not be used for strict predictions, but directionally, on average, should be valuable in assessing future points contributions from forwards. Also, while the point estimates from the models were rather accurate, one must exercise caution applying a group statistic to any individual (especially since the dataset size was only 160 observations).

Conclusions

Given the intuitive argument that a player slows down with age, we were able to put some math behind that assertion and, further, show that those players who started their NHL careers earlier in life generally produced and played more seasons. As noted in the *Pitfalls & Caveats* section above, the shapes of the curves represent group statistics and may not be applicable to any one individual player. They do, however, provide insight into the typical pattern of point production over time. This exercise served to validate the common conjecture regarding age (which might appear obvious) and wrap some numbers around that assumption. Those who started their NHL careers at an older age were predicted to peak earlier, as seen in the first derivative curves for the models. As well, they were estimated to produce less, percentage-wise, in their “twilight” years. Such information might be of value to a General Manager contemplating an offer to an older free agent and help answer the question, “How much has this player got left?”

Appendix

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For more information on hockey analytics, see Alan Ryder's web site: www.hockeyanalytics.com

or join the Hockey Analytics Group (HAG) at: http://sports.groups.yahoo.com/group/HAG_list/